# Data Communication & Computer Networks (CN)

## Topic 5 of 7 : Transmission Medium

**46 Pending Questions** — staged from repo exam PDFs & trusted sources — **NEED VERIFICATION** (answers + Hindi to be verified during topic build)

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)**

Total Questions: **46** pending questions (staging)

Question Type: MCQ — 4 options (A, B, C, D) per question, in English + Hindi

Layout: Left = **question** | Right = **explanation** (green): ✔ **why the correct answer is right** + ★ **Nukta** (key exam point)

Language rule: Hindi is the second language; **technical terms stay in English** (topology, protocol, jitter…)

Verification: Every answer cross-verified **4×** (Forouzan + official answer keys of Subject Test / Unit Test Solutions). Wrong options verified internally.

## Syllabus Coverage — sub-topic wise order

**How to read:** Left side: question + 4 options (try it yourself first). Right side (green): ✔ correct answer with why it is right, and ★ **Nukta** — the one-line exam tip to remember. Practice in 2 passes: first read question + options only, then check the green column.

**System:** generated by **tools/cn_build_topic.py** from **tools/data/topic5_data.py**. To add more questions later: drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions auto-classify topic-wise. See **Computer Network/README.md**.

## Pending Review — auto-ingested, needs verification

Auto-extracted & classified; answers to be verified before counting.

**P1.** "Manchester Encoding" and "Differential Manchester Encoding" are signal encoding techniques used at which layer? "Manchester Encoding" "Differential Manchester Encoding"

**(A)** Physical Layer
**(B)** Data Link Layer
**(C)** Network Layer
**(D)** Presentation Layer

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

**P2.** Assertion
**(A)** : Fiber Optic cable is immune to Electromagnetic Interference (EMI). (A): (EMI) Reason (R): It transmits data in the form of light pulses rather than electrical signals. (R): (A) Both (A) and (R) are true, and (R) is the correct explanation of (A)
**(B)** Both (A) and (R) are true, but (R) is NOT the correct explanation.
**(C)** (A) is true, but (R) is false. /.
**(D)** (A) is false, but (R) is true.
[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

**P3.** A company needs long-distance communication (5–10 km), high data rate, and immunity from electromagnetic interference (EMI). Which transmission media is most suitable? (5–10 ), EMI transmission media
**(A)** UTP
**(B)** Coaxial cable
**(C)** Single-mode fiber
**(D)** Multi-mode fiber
[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

**P4.** In a copper cable network, attenuation increases when frequency increases. What change will reduce attenuation? frequency attenuation attenuation
**(A)** Increase cable length
**(B)** Use lower category UTP
**(C)** Use thicker copper core
**(D)** Increase signal frequency
[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

**P5.** A network uses coaxial cable but suffers from signal reflection. Which problem is most likely the cause? coaxial cable signal reflection
**(A)** EMI
**(B)** Incorrect termination resistance
**(C)** Crosstalk
**(D)** Low bandwidth
[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

**P6.** Which of the following statements are TRUE for fiber optic cables? I. Immune to tapping II. High bandwidth III. Requires electrical grounding IV. Works using total internal reflection
**(A)** I, II, IV
**(B)** II, III
**(C)** I, III, IV
**(D)** I, II
[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

**P7.** A UTP cable used in Gigabit Ethernet shows high delay skew. Which is the cause? Gigabit Ethernet UTP delay skew

**(A)** Different twist rates in cable pairs
**(B)** Excessive EMI
**(C)** Fiber splice error
**(D)** Wrong wireless channel

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

**P8.** A channel has high bandwidth but suffers from high noise. Which encoding scheme will provide better error resistance? bandwidth noise - encoding error resistance

**(A)** NRZ
**(B)** Manchester
**(C)** AMI
**(D)** ASK

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

**P9.** Which phase of a computer virus life cycle is responsible for remaining hidden until a specific condition is met? D H

**(A)** Propagation Phase
**(B)** Triggering Phase
**(C)** Dormant Phase / C ( E )
**(D)** Execution Phase / F

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ auto-ingested — answer needs verification

**P10.** Reduction in signal strength over distance is called: v %– B

**(A)** Distortion –
**(B)** Noise –
**(C)** Attenuation – L
**(D)** Interference – T

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

**P11.** Which noise is caused due to electromagnetic coupling between nearby cables? ™S v - š›

**(A)** Thermal noise – Z
**(B)** Crosstalk – ]
**(C)** Impulse noise – ^_
**(D)** Quantization noise – c

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

**P12.** Which parameter of an analog signal represents the number of cycles per second? v ‰ ‡ 9

**(A)** Amplitude –
**(B)** Phase –
**(C)** Frequency – k
**(D)** Wavelength – pZ

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ auto-ingested — answer needs verification

**P13.** Which impairment cannot be completely eliminated but only minimized? Ÿ B y
**(A)** Attenuation – L
**(B)** Distortion –
**(C)** Noise –
**(D)** Delay –
[Source: BCI Subject Test-9.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

**P14.** Which of the following is a wireless networking technology? / H I + K
**(A)** Ethernet
**(B)** Wi-Fi
**(C)** Fiber Optic / H
**(D)** DSL
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

**P15.** What is the maximum data rate of a transmission medium known as? / ? Q
**(A)** Bandwidth / P
**(B)** Throughput / R
**(C)** Latency / T
**(D)** Jitter
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

**P16.** Which cable has maximum bandwidth? ?**
**(A)** Coaxial
**(B)** Twisted pair
**(C)** Optical fiber
**(D)** USB cable
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

**P17.** Which wireless standard provides the highest data rate
**(A)** 802.11b
**(B)** 802.11g
**(C)** 802.11n
**(D)** 802.11ac What is the purpose of DHCP Discover message?DHCP Discover (a) Request IP lease (b) Identify gateway (c) Detect MAC address (d) Announce server
[Source: CN Topic Test 6.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ auto-ingested — answer needs verification

**P18.** What kind of transmission medium is most appropriate to carry data in a computer network that is exposed to electrical interferences
**(A)** Unshielded twisted pair
**(B)** Optical fiber
**(C)** Coaxial cable
**(D)** Microwave
[Source: CN Topic Test-01.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

**P19.** Which of the following cannot be used as a medium for 802.3 ethernet? 802.3

**(A)** A thin coaxial cable
**(B)** A twisted pair cable
**(C)** A microwave link
**(D)** A fiber optical cable

[Source: CN Topic Test-01.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

**P20.** What technique is used to allow multiple signals to share the same communication channel in a network

**(A)** Multiplexing
**(B)** Phase modulation
**(C)** Analog modulation
**(D)** Digital modulation

[Source: CN Topic Test-02.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

**P21.** What is the meaning of bandwidth in a network

**(A)** Transmission capacity of a communication channel
**(B)** Connected components in a network
**(C)** Class of IP used in Network
**(D)** Interconnected by communication channels

[Source: CN Topic Test-02.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

**P22.** The period of a signal is 100ms. What is its frequency in kilohertz? 3. 100

**(A)** 10-1 kHz
**(B)** 10 -2 kHz
**(C)** 10-3 kHz
**(D)** 10-4 kHz

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ auto-ingested — answer needs verification

**P23.** Which of the following is NOT a type of transmission media

**(A)** Coaxial cable
**(B)** Fiber optics
**(C)** NIC
**(D)** Twisted pair

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

**P24.** In digital communication, what does "attenuation" refer to? " "

**(A)** Signal delay
**(B)** Signal loss
**(C)** Signal interference
**(D)** Signal amplification

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

**P25.** Which of the following best explains why fiber optic cables have a lower bit error rate than copper cables

**(A)** Higher signal power
**(B)** Immunity to electromagnetic interference
**(C)** Simpler modulation techniques
**(D)** Larger physical size

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

**P26.** Given a bit rate of b bits/sec, the time required to send 16 bits is: b / , 16 required :

**(A)** 16*b sec
**(B)** 16/b sec
**(C)** 16b sec
**(D)** b16 sec

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

**P27.** Which of the following transmission media works on the principle of total internal reflection

**(A)** Optical fiber cable
**(B)** Shielded twisted pair cable
**(C)** Unshielded twisted pair cable
**(D)** coaxial cable

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

**P28.** An analog signal carries 4 bits in each signal unit. if 1000 signal units are sent per second, then baud rate and bit rate of the signal are 4 1000

**(A)** 1000 bauds/sec, 4000 bps
**(B)** 1000 bauds/sec, 500 bps
**(C)** 4000 bauds/sec, 1000 bps
**(D)** 2000 bauds/sec, 1000 bps

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

**P29.** Crosstalk is typically minimized in twisted pair cables by:

**(A)** Increasing the power level
**(B)** Increasing the twist rate
**(C)** Adding shielding to the core
**(D)** Using longer cables

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

**P30.** What is the typical maximum length of an unshielded twisted pair (UTP) cable in Ethernet before signal degradation occurs? ( )

**(A)** 50 meters
**(B)** 100 meters
**(C)** 150 meters
**(D)** 200 meters

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

**P31.** Which connector is commonly used for coaxial cable in physical networks

**(A)** RJ-45
**(B)** USB-C
**(C)** BNC
**(D)** HDMI

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

**P32.** Which type of wireless transmission requires line-of-sight

**(A)** Wi-Fi
**(B)** Infrared
**(C)** Radio
**(D)** Bluetooth

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ auto-ingested — answer needs verification

**P33.** The effective bandwidth of a communication channel is primarily limited by:

**(A)** Protocol overhead
**(B)** Noise and attenuation
**(C)** File size
**(D)** MAC address filtering

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

**P34.** Which of the following statements is not true with respect to microwaves

**(A)** Electromagnetic waves with frequencies from 300 GHz to 400 Thz.
**(B)** Propagation is line-of-sight.
**(C)** Very high-frequency waves cannot penetrate walls.
**(D)** Use of certain portions of the band requires permission from authorities. 28 Optical fiber uses reflection to guide light through a channel, in which angle of incidence is_____the critical angle. _____ (a) A equal to (b) less than (c) greater than (d) less than or equal to

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ auto-ingested — answer needs verification

**P35.** Which of the following transmission media is used in Bluetooth Technology: :

**(A)** Radio links
**(B)** Microwave links
**(C)** CVSAT Communication
**(D)** Fiber-optic

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ auto-ingested — answer needs verification

**P36.** The primary use of scramblers in Line Coding Schemes is to -
**(A)** make the data look random
**(B)** compress transmission data
**(C)** increase signal bandwidth
**(D)** decrease signal bandwidth
[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

**P37.** - transmission medium modern internet backbone networks primary ? Which transmission medium is primarily used in modern internet backbone networks
**(A)** Twisted Pair Cable
**(B)** Coaxial Cable
**(C)** Single Mode Fiber Optic Cable
**(D)** Microwave Links
[Source: BCI Subject Test-8.pdf | auto-classified: Topic 1 / DC Components & Protocol Basics]

⚠ auto-ingested — answer needs verification

**P38.** Which physical layer technique allows multiple users to share the same bandwidth simultaneously
**(A)** TDMA
**(B)** FDMA
**(C)** CDMA
**(D)** ARQ
[Source: CN Topic Test-04.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

**P39.** signals are discrete and generated by digital modulations. Analog Digital Noise Bandwidth
**(A)** T
**(B)** n
[Source: Random Test- Rajasthan PYQ.pdf (OCR), p5 | auto-classified: Topic 5 / Signals & Performance]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

**P40.** CQuanizing 39. Quantizing noise occurs in Pulbre Gade Moduoechon APCM
**(A)** TDM
**(B)** FDM
**(C)** PPM
**(D)** PCM (B ) TDM
[Source: Random Test- Rajasthan PYQ.pdf (OCR), p10 | auto-classified: Topic 5 / Signals & Performance]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

**P41.** Llhh Division Mwiple Access 83. Select the false statement about CDMA technology 益益 I.CDMA technology optimizes theuseof availablebandwidthasit transmitsover the C1:SL entire frequency range. (2:S2 ≤s:≤) II.Each user in a CDMA system uses a c4:54 different code to modulate their signal y 43 CDMA technology offers lesssecurity as compared to theGSM technology. A SIMcard is alwaysrequired for the Gmpvice workingofa CDMA device. CDMA Moc eCM Only (I) Xonly (H) Io GSM ()and (II) (Ill) and (IV)
[Source: Random Test- Rajasthan PYQ.pdf (OCR), p11 | auto-classified: Topic 5 / Unguided Media]

⚠ OCR-parse staging: verify options/answer at curation (see src page). OPTIONS INCOMPLETE in OCR — re-check source page.

**P42.** 45.If the channel band is limited to 6kHz and channe signal to noise ratio is 16,what would be the capacity of channel? og(4 zgir

**(A)** 15.15kbps 24.74kbps Radio Noicy Nosdess (g(8)
**(B)** 30.12kbps
**(C)** 52.18kbps chann channel (dNs+) 01009 (1+16)
**(D)** 15.15kbps

[Source: Random Test- Rajasthan PYQ.pdf (OCR), p30 | auto-classified: Topic 5 / Signals & Performance]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

**P43.** 8. Which of the ‘following Ethernet encoding has the highest redundancy' Encading ) 1B2B (q) 4B5B 5B6B prerent g< 8B10B abser. Fke Nw ophcad 3J59

[Source: Unit Test-4 Solution.pdf (OCR), p9 | auto-classified: Topic 5 / Signals & Performance]

⚠ OCR-parse staging: verify options/answer at curation (see src page). OPTIONS INCOMPLETE in OCR — re-check source page.

**P44.** 22. The amount of data transmitted in a fixed amount of time iscalled

**(A)** Refresh rate (b) Resolution efBandwidth
**(B)** Frequency

[Source: Unit Test-4 Solution.pdf (OCR), p23 | auto-classified: Topic 5 / Unguided Media]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

**P45.** 29. The encoding technique used to transmit the signal in giga Ethernet technology over fiber optic medium is

**(A)** Differential Manchester encoding
**(B)** Non Return to zero
**(C)** 4B
**(D)** 8B

[Source: Unit Test-4 Solution.pdf (OCR), p30 | auto-classified: Topic 5 / Signals & Performance]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

**P46.** Communication using a modulated carrier to apply a data signal to a transmission medium in analog form is:

**(A)** Broadband networking
**(B)** Baseband networking
**(C)** Modem
**(D)** Local area network

[Source: IBPS RRB PO Computer Awareness PYQ (via Meritnotes) | auto-classified: Topic 5 / DC Components & Protocol Basics]

⚠ Site key: broadband uses modulated analog carrier; baseband is digital.